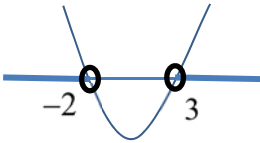
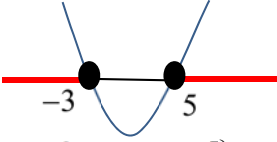


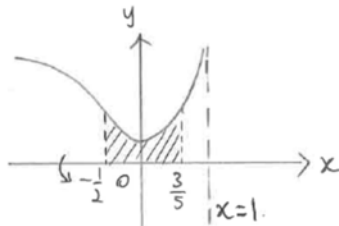
PU3 MATHEMATICS

Qn.	Suggested Solution
1	From MF26, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots$ $= \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^r}{r!} + \dots$ $= \sum_{r=0}^{\infty} \frac{x^r}{r!}$ $\sum_{r=2}^{\infty} \frac{3^r}{r!} = \sum_{r=0}^{\infty} \frac{3^r}{r!} - \sum_{r=0}^1 \frac{3^r}{r!}$ $= \sum_{r=0}^{\infty} \frac{3^r}{r!} - \frac{3^0}{0!} - \frac{3^1}{1!}$ $= e^3 - 4$
2	$\frac{x-7}{x^2-x-6} \leq 1$ $\frac{x-7-(x^2-x-6)}{x^2-x-6} \leq 0$ $\frac{-x^2+2x-1}{x^2-x-6} \leq 0$ $\frac{x^2-2x+1}{x^2-x-6} \geq 0$ $\frac{(x-1)^2}{(x+2)(x-3)} \geq 0$ For $(x-1)^2 > 0$, $\frac{1}{(x+2)(x-3)} > 0$ $(x+2)(x-3) > 0$ $x < -2 \quad \text{or} \quad x > 3$  For $(x-1)^2 = 0$, $x = 1$ $\therefore x < -2 \quad \text{or} \quad x > 3 \quad \text{or} \quad x = 1$ $\frac{\ln x - 7}{(\ln x)^2 - \ln x - 6} \leq 1$ Replacing x with $\ln x$: $\ln x < -2 \quad \text{or} \quad \ln x > 3 \quad \text{or} \quad \ln x = 1$ $0 < x < e^{-2} \quad \text{or} \quad x > e^3 \quad \text{or} \quad x = e$

3(i)	Graph of $y = \frac{1}{f(x)}$ showing a vertical asymptote at $x=a$ and a horizontal asymptote at $y = \frac{1}{k}$. The curve passes through the origin $(0,0)$ and a point $(b, \frac{1}{c})$.
3(ii)	$y = f'(x)$ Graph of $y = f'(x)$ showing a vertical asymptote at $x=0$ and a horizontal asymptote at $y=0$. The curve passes through the origin $(0,0)$ and a point $(b,0)$.
4(i)	$y = \frac{2x^2 + 13x + 23}{x + \lambda}$ $= 2x + 13 - 2\lambda + \frac{23 - 13\lambda + 2\lambda^2}{x + \lambda}$ The asymptotes are $y = 2x + 13 - 2\lambda$ and $x = -\lambda$ When $x = 0$, $y = \frac{23}{\lambda}$ When $y = 0$, $\frac{2x^2 + 13x + 23}{x + \lambda} = 0$ $2x^2 + 13x + 23 = 0$ Using GC, there are no real roots $\Rightarrow$ no x-intercepts Graph of $y = \frac{2x^2 + 13x + 23}{x + \lambda}$ showing a vertical asymptote at $x = -\lambda$ and a slant asymptote $y = 2x + 13 - 2\lambda$. The curve passes through the point $(0, \frac{23}{\lambda})$.
4(ii)	Method 1: Discriminant

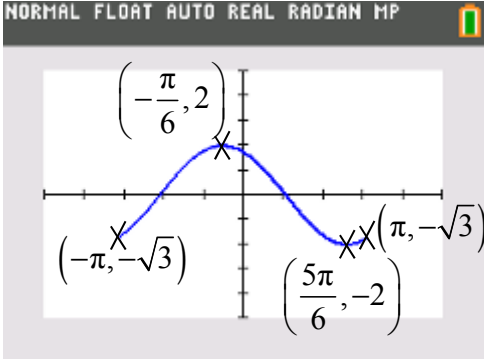
	$y = \frac{2x^2 + 13x + 23}{x + 3}$ $y(x + 3) = 2x^2 + 13x + 23$ $2x^2 + (13 - y)x + 23 - 3y = 0$ To determine the range of values of that y can take : For real roots, $b^2 - 4ac \geq 0$ $(13 - y)^2 - 4(2)(23 - 3y) \geq 0$ $y^2 - 2y - 15 \geq 0$ $(y + 3)(y - 5) \geq 0$ $y \leq -3 \text{ or } y \geq 5 \Rightarrow \{y \in \mathbb{R} : y \leq -3 \text{ or } y \geq 5\}$  Method 2: Differentiation $y = \frac{2x^2 + 13x + 23}{x + 3}$ $\frac{dy}{dx} = \frac{(x + 3)(4x + 13) - (2x^2 + 13x + 23)}{(x + 3)^2}$ $= \frac{4x^2 + 25x + 39 - 2x^2 - 13x - 23}{(x + 3)^2}$ $= \frac{2x^2 + 12x + 16}{(x + 3)^2}$ To find turning points, $\frac{dy}{dx} = 0$ $\frac{2x^2 + 12x + 16}{(x + 3)^2} = 0$ $2x^2 + 12x + 16 = 0$ $x^2 + 6x + 8 = 0$ $(x + 4)(x + 2) = 0$ $\therefore x = -4 \text{ or } -2$ When $x = -4$, $y = -3$, When $x = -2$, $y = 5$ $\therefore y \leq -3 \text{ or } y \geq 5 \Rightarrow \{y \in \mathbb{R} : y \leq -3 \text{ or } y \geq 5\}$
5(i)	Method 1: Implicit differentiation $\ln y = \sin 2x$ Differentiating both sides with respect to x: $\frac{1}{y} \frac{dy}{dx} = 2 \cos 2x$ $\frac{dy}{dx} = 2y \cos 2x \text{ (shown)}$

	<u>Method 2: Explicit differentiation</u> $\ln y = \sin 2x$ $y = e^{\sin 2x}$ $\frac{dy}{dx} = 2 \cos 2x e^{\sin 2x}$ $\frac{dy}{dx} = 2y \cos 2x$ (shown) Differentiating both sides with respect to x: $\frac{d^2 y}{dx^2} = 2 \cos 2x \frac{dy}{dx} - 4y \sin 2x$ When $x = 0$, $y = 1$, $\frac{dy}{dx} = 2$, $\frac{d^2 y}{dx^2} = 4$
5(ii)	$y = 1 + 2x + 4\left(\frac{x^2}{2!}\right) + \dots = 1 + 2x + 2x^2 + \dots$
5(iii)	$e^{px} \ln(1 + qx)$ $= \left(1 + px + \frac{(px)^2}{2!} + \dots\right) \left(qx - \frac{(qx)^2}{2} + \dots\right)$ $= \left(1 + px + \frac{p^2 x^2}{2} + \dots\right) \left(qx - \frac{q^2 x^2}{2} + \dots\right)$ $= qx - \frac{q^2 x^2}{2} + pqx^2 + \dots$ $= qx + \left(pq - \frac{q^2}{2}\right)x^2 + \dots$ By comparison, $q = 2, pq - \frac{q^2}{2} = 2 \Rightarrow p = 2$ The expansion is valid for $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$

6(i)	$x = 1 - e^{-t}, \quad y = 1 + t^2$ $\frac{dx}{dt} = e^{-t}, \quad \frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = 2t \cdot \frac{1}{e^{-t}} = 2te^t$ When $t = 1$, $\frac{dy}{dx} = 2e$ The required gradient is $2e$
6(ii)	$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx} = (2te^t + 2e^t) \cdot \frac{1}{e^{-t}} = 2e^{2t}(t+1)$ Since $2e^{2t} > 0$ for $t \in \mathbb{R}$ but $t+1 > 0$ only for $t > -1$, $\frac{d^2y}{dx^2} = 2e^{2t}(t+1) > 0 \text{ only for } t > -1.$ $\therefore$ The curve is not concave upwards for all real values of t. Note: $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx}$
6(iii)	$x = 1 - e^{-t} \Rightarrow e^{-t} = 1 - x \Rightarrow t = -\ln(1 - x)$ Substituting $t = -\ln(1 - x)$ into $y = 1 + t^2$, $y = 1 + [\ln(1 - x)]^2$
6(iv)	The required volume $= \pi \int_{-\frac{1}{2}}^{\frac{3}{5}} (1 + [\ln(1 - x)]^2)^2 dx$ $= 4.6699 = 4.670 \text{ units}^3 \text{ (3 d.p.)}$ 

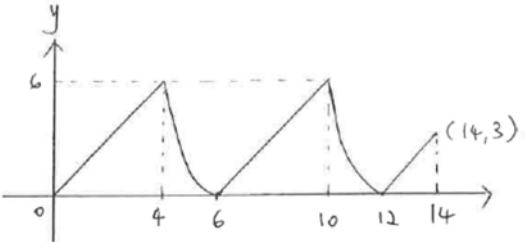
7(a)(i)	$\int \cos 2x \sin^5 2x \, dx = \frac{1}{2} \int 2 \cos 2x \sin^5 2x \, dx$ $= \frac{1}{2} \cdot \frac{\sin^{5+1} 2x}{5+1} + c$ $= \frac{1}{12} \sin^6 2x + c$
7(a)(ii)	$\int \frac{e^x}{\sqrt{1-2e^{2x}}} \, dx$ $u = e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow \frac{du}{dx} = u$ Method 1 $\int \frac{e^x}{\sqrt{1-2e^{2x}}} \, dx = \int \frac{u}{\sqrt{1-2u^2}} \cdot \frac{1}{u} \, du$ $= \int \frac{1}{\sqrt{1-2u^2}} \, du$ $= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\frac{1}{2}-u^2}} \, du$ $= \frac{1}{\sqrt{2}} \int \frac{1}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 - u^2}} \, du$ $= \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2}u) + c$ $= \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2}e^x) + c$ Method 2 $\int \frac{e^x}{\sqrt{1-2e^{2x}}} \, dx = \int \frac{u}{\sqrt{1-2u^2}} \cdot \frac{1}{u} \, du$ $= \int \frac{1}{\sqrt{1-2u^2}} \, du$ $= \frac{1}{\sqrt{2}} \int \frac{\sqrt{2}}{\sqrt{1-(\sqrt{2}u)^2}} \, du$ $= \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2}u) + c$ $= \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2}e^x) + c$

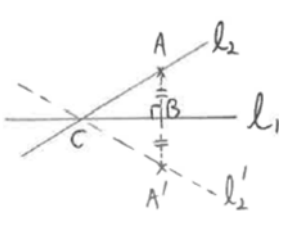
7(b)(i)	$\int x^2 \sin nx \, dx$ $= x^2 \left(-\frac{\cos nx}{n} \right) + \frac{2}{n} \int x \cos nx \, dx$ $= -\frac{x^2}{n} \cos nx + \frac{2}{n} \left[x \left(\frac{\sin nx}{n} \right) - \frac{1}{n} \int \sin nx \, dx \right]$ $= -\frac{x^2}{n} \cos nx + \frac{2}{n} \left[\frac{x}{n} \sin nx + \frac{1}{n^2} \cos nx \right] + c$ $= \frac{2}{n^3} \cos nx + \frac{2x}{n^2} \sin nx - \frac{x^2}{n} \cos nx + c$
7(b)(ii)	$\int_{\pi}^{2\pi} x^2 \sin nx \, dx$ $= \left[\frac{2}{n^3} \cos nx - \frac{x^2}{n} \cos nx + \frac{2x}{n^3} \sin nx \right]_{\pi}^{2\pi}$ $= \frac{2}{n^3} \cos 2\pi n - \frac{(2\pi)^2}{n} \cos 2\pi n + \frac{2}{n^3} (2\pi) \sin 2\pi n - \left[\frac{2}{n^3} \cos \pi n - \frac{\pi^2}{n} \cos \pi n + \frac{2}{n^3} (\pi) \sin \pi n \right]$ $= \frac{2}{n^3} (1) - \frac{4\pi^2}{n} (1) + 0 - \left[\frac{2}{n^3} \cos \pi n - \frac{\pi^2}{n} \cos \pi n + 0 \right]$ $= \frac{2}{n^3} - \frac{4\pi^2}{n} - \left(\frac{2}{n^3} - \frac{\pi^2}{n} \right) \cos \pi n$ $\cos \pi n = \begin{cases} -1 & \text{for } n \text{ odd} \\ 1 & \text{for } n \text{ even} \end{cases}$ For n odd, $\frac{2}{n^3} - \frac{4\pi^2}{n} - \left(\frac{2}{n^3} - \frac{\pi^2}{n} \right) \cos \pi n$ $= \frac{2}{n^3} - \frac{4\pi^2}{n} - \left(\frac{2}{n^3} - \frac{\pi^2}{n} \right) (-1)$ $= \frac{4}{n^3} - \frac{5\pi^2}{n} \Rightarrow p = 4, q = 5$ For n even, $\frac{2}{n^3} - \frac{4\pi^2}{n} - \left(\frac{2}{n^3} - \frac{\pi^2}{n} \right) \cos \pi n$ $= \frac{2}{n^3} - \frac{4\pi^2}{n} - \left(\frac{2}{n^3} - \frac{\pi^2}{n} \right) (1)$ $= -\frac{3\pi^2}{n} \Rightarrow p = 0, q = 3$

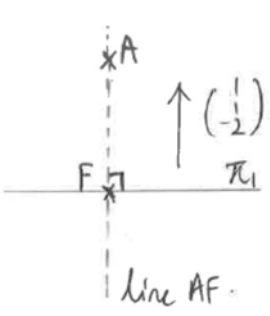
	$\therefore \int_{\pi}^{2\pi} x^2 \sin nx \, dx = \begin{cases} \frac{4}{n^3} - \frac{5\pi^2}{n} & \text{for } n \text{ odd} \\ -\frac{3\pi^2}{n} & \text{for } n \text{ even} \end{cases}$
8(i)	$R \cos \alpha = \sqrt{3} \quad ; \quad R \sin \alpha = 1$ $R^2 = \sqrt{3}^2 + 1^2 \Rightarrow R = 2 \, (R > 0)$ $\tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6} \, (\alpha \text{ is acute})$ $f(x) = \sqrt{3} \cos x - \sin x = 2 \cos \left(x + \frac{\pi}{6} \right)$
8(ii)	 A horizontal line $y = k$, $k \in (-2, 2)$ intersects the graph of $y = f(x)$ at more than one point. f is not a one-one function. f^{-1} does not exist.
8(iii)	From the graph, the maximum value of k is $\frac{5\pi}{6}$
8(iv)	Let $y = f(x)$ $y = 2 \cos \left(x + \frac{\pi}{6} \right)$ $x = \cos^{-1} \left(\frac{y}{2} \right) - \frac{\pi}{6}$ $f^{-1}(y) = \cos^{-1} \left(\frac{y}{2} \right) - \frac{\pi}{6}$ $f^{-1}(x) = \cos^{-1} \left(\frac{x}{2} \right) - \frac{\pi}{6}$ Domain of $f^{-1} = \text{Range of } f = [-2, 2]$
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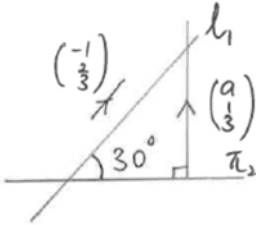
	Endpoints for $y = f^{-1}f(x) : A\left(-\frac{\pi}{6}, -\frac{\pi}{6}\right), B\left(\frac{5\pi}{6}, \frac{5\pi}{6}\right)$ Endpoints for $y = f(x) : C\left(-\frac{\pi}{6}, 2\right), D\left(\frac{5\pi}{6}, -2\right)$ Endpoints for $y = f^{-1}(x) : E\left(-2, \frac{5\pi}{6}\right), F\left(2, -\frac{\pi}{6}\right)$
8(v)	$ \begin{aligned} g^{2018}(2) &= g^3 g^{2015}(2) \\ &= g^{2015}(2) \\ &= g^3 g^{2012}(2) \\ &= g^{2012}(2) \\ &= \dots \\ &= g^3 g^2(2) \\ &= g^2(2) \\ &= \frac{2-1}{2} = \frac{1}{2} \end{aligned} $
9(i)	$ \begin{aligned} T_n &= S_n - S_{n-1} \\ &= \frac{3}{2}n^2 + \frac{13}{2}n - \left[\frac{3}{2}(n-1)^2 + \frac{13}{2}(n-1) \right] \\ &= \frac{3}{2}n^2 + \frac{13}{2}n - \left[\frac{3}{2}n^2 - 3n + \frac{3}{2} + \frac{13}{2}n - \frac{13}{2} \right] \\ &= 3n + 5 \end{aligned} $
9(ii)	The common ratio, $r = 0.8$ Since $-1 < r < 1$, the sequence is convergent
9(iii)	$S_{\infty} = \frac{5}{1-0.8} = 25 < 35$ $\therefore$ Terra will not reach the peak of the hill

9(iv)	<u>Method 1</u> T_1, T_2, T_3, T_4, T_5 $5, 5(0.8), 5(0.8^2), 5(0.8^3), 5(0.8^3)(0.9)$ $U_1 = 5(0.8^3)(0.9), U_2, \dots, U_n$ $T_5 = U_1$ i.e. T_5 is the first term of the GP after r is changed from 0.8 to 0.9. $(T_1 + T_2 + T_3 + T_4) + (U_1 + U_2 + U_3 + \dots + U_n) = 35$ $\frac{5(1-0.8^4)}{1-0.8} + \frac{5(0.8^3)(0.9)(1-0.9^n)}{1-0.9} = 35$ $\frac{5(0.8^3)(0.9)(1-0.9^n)}{1-0.9} = 20.24$ $1-0.9^n = 0.87847$ $0.9^n = 0.12153$ $n = \frac{\ln 0.12153}{\ln 0.9} = 20.004$ The minimum no. of 10-minute periods required $= 4 + 21 = 25$ <u>Method 2</u> T_1, T_2, T_3, T_4 $5, 5(0.8), 5(0.8^2), 5(0.8^3)$ $U_1 = 5(0.8^3), U_2 = 5(0.8^3)(0.9), \dots, U_n$ $T_4 = U_1$ i.e. T_4 is the first term of the GP. $(T_1 + T_2 + T_3) + (U_1 + U_2 + U_3 + \dots + U_n) = 35$ $\frac{5(1-0.8^3)}{1-0.8} + \frac{5(0.8^3)(1-0.9^n)}{1-0.9} = 35$ $1-0.9^n = 0.890625$ $0.9^n = 0.109375$ $n = \frac{\ln 0.109375}{\ln 0.9} = 21.004$ The minimum number of 10-minute periods required $= 3 + 22$ $= 25$
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9(v)	
9(vi)	<u>Method 1</u> The distance covered $= \int_4^6 \frac{3}{2}t^2 - 18t + 54 \, dt + \frac{1}{2}(4)(6) + \int_4^5 \frac{3}{2}t^2 - 18t + 54 \, dt$ $= 19.5 \text{ m}$ <u>Method 2</u> The distance covered $= \int_4^6 \frac{3}{2}t^2 - 18t + 54 \, dt + \frac{1}{2}(4)(6) + \int_4^5 \frac{3}{2}t^2 - 18t + 54 \, dt$ $= \left(\frac{1}{2}t^3 - 9t^2 + 54t \right)_4^6 + 12 + \left(\frac{1}{2}t^3 - 9t^2 + 54t \right)_4^5$ $= (108 - 104) + 12 + \left(\frac{215}{2} - 104 \right) = 19\frac{1}{2} \text{ m}$
10(i)	$l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + \alpha \begin{pmatrix} -1 \\ 6 \\ 12 \end{pmatrix}, \alpha \in \mathbb{R}$

	Equating the equations of both lines, $\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + \alpha \begin{pmatrix} -1 \\ 6 \\ 12 \end{pmatrix}$ $\begin{pmatrix} -2 - \lambda \\ 1 + 2\lambda \\ 3 + 3\lambda \end{pmatrix} = \begin{pmatrix} -\alpha \\ 5 + 6\alpha \\ 15 + 12\alpha \end{pmatrix}$ $2 + \lambda = \alpha \Rightarrow \alpha - \lambda = 2$ $1 + 2\lambda = 5 + 6\alpha \Rightarrow 3\alpha - \lambda = -2$ Solving both equations simultaneously, $\alpha = -2, \lambda = -4$ $\vec{OC} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + (-4) \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -7 \\ -9 \end{pmatrix}$
10(ii)	By Ratio Theroem, $\vec{OB} = \frac{\vec{OA} + \vec{OA'}}{2}$ $\begin{pmatrix} -5 \\ 7 \\ 12 \end{pmatrix} = \frac{1}{2} \left[\begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + \vec{OA'} \right]$ $\vec{OA'} = \begin{pmatrix} -10 \\ 9 \\ 9 \end{pmatrix}$ $\vec{CA'} = \begin{pmatrix} -10 \\ 9 \\ 9 \end{pmatrix} - \begin{pmatrix} 2 \\ -7 \\ -9 \end{pmatrix} = \begin{pmatrix} -12 \\ 16 \\ 18 \end{pmatrix} = 2 \begin{pmatrix} -6 \\ 8 \\ 9 \end{pmatrix}$ 

	<u>Method 1</u> A vector equation of the line of intersection is $\mathbf{r} = \begin{pmatrix} -10 \\ 9 \\ 9 \end{pmatrix} + s \begin{pmatrix} -6 \\ 8 \\ 9 \end{pmatrix}, s \in \mathbb{R}$ Cartesian equation of the line of intersection is $\frac{-x-10}{6} = \frac{y-9}{8} = \frac{z-9}{9}$ <u>Method 2</u> A vector equation of the line of intersection is $\mathbf{r} = \begin{pmatrix} 2 \\ -7 \\ -9 \end{pmatrix} + t \begin{pmatrix} -6 \\ 8 \\ 9 \end{pmatrix}, t \in \mathbb{R}$ Cartesian equation of the line of intersection is $\frac{2-x}{6} = \frac{y+7}{8} = \frac{z+9}{9}$
10(iii)	$x + y - 2z = -1 \Leftrightarrow \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = -1$ Let the point F be the foot of the perpendicular from A to π_1 Line AF: $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \mu \in \mathbb{R}$ $\pi_1: x + y - 2z = -1 \Leftrightarrow \mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = -1$ By substitution, $\left[\begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right] \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = -1$ $\begin{pmatrix} \mu \\ 5 + \mu \\ 15 - 2\mu \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = -1$ $6\mu - 25 = -1 \Rightarrow \mu = 4$ 

	$\vec{OF} = \begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} + 4 \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \\ 7 \end{pmatrix} \Rightarrow F \text{ is } (4, 9, 7)$ $\vec{AF} = \begin{pmatrix} 4 \\ 9 \\ 7 \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \\ 15 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix}$ The required perpendicular distance $= \vec{AF} = \sqrt{4^2 + 4^2 + (-8)^2} = \sqrt{96} = 4\sqrt{6} \text{ units}$
10(iv)	$\sin 30^\circ = \frac{\begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} a \\ 1 \\ 3 \end{pmatrix}}{\sqrt{14}\sqrt{a^2+10}}$ $\left(\frac{1}{2}\right)\sqrt{14}\sqrt{a^2+10} = -a+11$ $\sqrt{14(a^2+10)} = 2(11-a)$ $14(a^2+10) = 4(11-a)^2$ $10a^2 + 88a - 344 = 0$ $a = 2.9321 \text{ or } -11.732$ $a = 2.93 \text{ or } -11.7 \text{ (3 s.f.)}$ 
10(v)	$\pi_2 : ax + y + 3z = 5 \Leftrightarrow \text{rg} \begin{pmatrix} a \\ 1 \\ 3 \end{pmatrix} = 5$ A direction vector parallel to the line of intersection $= \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} a \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2a-3 \\ 1-a \end{pmatrix}$ By comparison to $\begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$, $-2a-3 = -1 \Rightarrow a = -1$ (verified by $1-a = 2 \Rightarrow a = -1$)

	$\pi_1: x + y - 2z = -1$ $\pi_2: -x + y + 3z = 5$ Let $z = 0$, $x + y = -1$, $-x + y = 5$ Solving both equations simultaneously, $y = 2$, $x = -3$. $(-3, 2, 0)$ lies on the line of intersection. A possible position vector of V is $\begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}$.
11(i)	$x_{n+1} - x_n = \frac{13}{2} - n$ $\sum_{r=1}^n (x_r - x_{r-1}) = \sum_{r=1}^n \left(\frac{13}{2} - r \right)$ $\begin{pmatrix} x_1 - x_0 \\ +x_2 - x_1 \\ +x_3 - x_2 \\ \dots \\ +x_{n-1} - x_{n-2} \\ +x_n - x_{n-1} \end{pmatrix} = \sum_{r=1}^n \frac{13}{2} - \sum_{r=1}^n r$ $x_n - x_0 = \frac{13}{2}n - \frac{n}{2}(n+1)$ $x_n - 8 = 6n - \frac{1}{2}n^2$ $x_n = -\frac{1}{2}n^2 + 6n + 8$
11(ii)	$\frac{dx}{dt} = k(50 - 2t - x)$ Given that $\frac{dx}{dt} = 14$, $x = 8$ when $t = 0$, $14 = k(50 - 0 - 8)$ $k = \frac{1}{3}$ $\frac{dx}{dt} = \frac{1}{3}(50 - 2t - x) \text{ (shown)}$

$$\frac{dx}{dt} = \frac{1}{3}(50 - 2t - x)$$

$$u = 2t + x \Rightarrow x = u - 2t$$

$$\frac{dx}{dt} = \frac{du}{dt} - 2$$

$$\frac{du}{dt} - 2 = \frac{1}{3}[50 - 2t - (u - 2t)]$$

$$\frac{du}{dt} = \frac{1}{3}(50 - u) + 2$$

$$\frac{du}{dt} = \frac{1}{3}(56 - u)$$

$$\frac{du}{dt} = \frac{1}{3}(56 - u)$$

$$\frac{1}{56 - u} \frac{du}{dt} = \frac{1}{3}$$

$$\int \frac{1}{56 - u} \frac{du}{dt} dt = \int \frac{1}{3} dt$$

$$-\ln|56 - u| = \frac{1}{3}t + C_1$$

$$\ln|56 - u| = -\frac{1}{3}t + C_2 \text{ where } C_2 = -C_1$$

$$|56 - u| = e^{-\frac{1}{3}t + C_2}$$

$$56 - u = \pm e^{C_2} e^{-\frac{1}{3}t}$$

$$56 - u = Ae^{-\frac{1}{3}t} \text{ where } A = \pm e^{C_2}$$

$$u = 56 - Ae^{-\frac{1}{3}t}$$

$$2t + x = 56 - Ae^{-\frac{1}{3}t}$$

$$x = 56 - Ae^{-\frac{1}{3}t} - 2t$$

Given that $x = 8$ when $t = 0$,

$$8 = 56 - Ae^{-\frac{1}{3}(0)} - 2(0) \Rightarrow A = 48$$

$$x = 56 - 48e^{-\frac{1}{3}t} - 2t$$

11(iii)	$\frac{d^2x}{dt^2} = -\frac{100}{(t+1)^3}$ $\int \frac{d^2x}{dt^2} dt = -100 \int (t+1)^{-3} dt$ $\frac{dx}{dt} = 50(t+1)^{-2} + C$ $\int \frac{dx}{dt} dt = 50 \int (t+1)^{-2} dt$ $x = -50(t+1)^{-1} + Ct + D$ $x = -\frac{50}{t+1} + Ct + D$ Given that $\frac{dx}{dt} = 41\frac{2}{3} = \frac{125}{3}$, $x = 8$ when $t = 0$, $\frac{125}{3} = 50(0+1)^{-2} + C \Rightarrow C = -\frac{25}{3}$ $8 = -\frac{50}{0+1} + 0 + D \Rightarrow D = 58$ $x = -\frac{50}{t+1} - \frac{25}{3}t + 58$
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